# Courier Management System (CMS) - Salesforce Implementation

## Topic

**Courier Management System (CMS) - A Salesforce-Based Logistics Automation Platform**

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## Team Members

| Name | Role | Responsibility Area |
|---|---|---|
| Sanjai S | Team Lead | Overall project coordination, Salesforce configuration, Flow automation |
| Sarathi M | Member | Object modeling, field design, page layouts, validation rules |
| Balakrishnan V | Member | Security (profiles & roles), reports, dashboards, and testing |

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## Overview

Courier and logistics companies frequently struggle to manage shipments and deliveries efficiently because they rely on manual processes and disconnected systems. Without a centralized tracking mechanism, it becomes difficult to monitor parcel movement in real time, which results in delayed delivery updates and poor visibility for customers. Delivery agent assignments are often performed manually, increasing the risk of scheduling conflicts and operational delays. The absence of automated notifications prevents timely communication with customers, and the lack of performance analytics limits managerial insight into operational efficiency. As companies expand across multiple branches, coordination becomes even more complex without an integrated platform - ultimately leading to customer dissatisfaction, reduced productivity, and revenue loss.

The **Courier Management System (CMS)** was built to directly address these challenges. It is a Salesforce-based cloud solution that centralizes shipment booking, real-time tracking, delivery agent assignment, automated status updates, billing, and performance analytics into a single unified platform. By replacing manual registers and fragmented spreadsheets with an integrated digital solution, CMS

improves operational visibility, strengthens customer communication, reduces delays and errors, and supports faster, data-driven decision-making.

**Problem Statements Identified:**

- *Branch Manager:* Struggles to manage shipments efficiently because tracking is manual and systems are disconnected - resulting in stress and operational conflicts.
- *Customer:* Struggles to track parcels because updates are delayed and there is no centralized system - resulting in uncertainty and dissatisfaction.

**Why Salesforce was chosen:** Among the alternatives considered (manual registers, spreadsheet tracking, a standalone portal, and Salesforce CRM), Salesforce was selected for its automation capabilities, scalability, and real-time visibility - all critical requirements for a growing, multi-branch courier operation.

**Category:** Logistics **Project Type:** Group Project **Skills Applied:** Salesforce Administration, Data Modelling, Flow Designer & Process Automation, Data Validation, Data Security (Profiles & Roles), Reports, Dashboards

**Technology Stack:**

| Layer | Technology Used |
|---|---|
| User Interface | Salesforce Lightning |
| Business Logic | Salesforce Flows |
| Database | Salesforce Custom Objects |
| Automation | Validation Rules, Record-Triggered Flows |
| Security | Profiles, Roles |
| Reporting | Custom Reports & Dashboards |

**Functional Requirements Addressed:**

| ID | Requirement |
|---|---|
| FR-1 | Customer management |
| FR-2 | Shipment booking |
| FR-3 | Delivery agent assignment |
| FR-4 | Real-time status updates |
| FR-5 | Invoice generation |
| FR-6 | Reports & dashboards |

**Non-Functional Requirements Addressed:** Usability (a clean Lightning interface), Security (role-based access control), Reliability (error-free automation via validation rules), Performance (fast tracking updates), Availability (24/7 cloud access), and Scalability (built to support growing branch networks).

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# Steps

## Milestone 1 - Salesforce Developer Account Setup

A Salesforce Developer Edition organization was created as the foundation for the entire project. This involved registering at developer.salesforce.com with organizational details (name, email, job title as “Developer,” company as the college name, and country as India), verifying the account through a confirmation email, and setting a secure password along with a security question. Upon verification, the team gained access to the Salesforce Setup environment where all subsequent configuration work was performed.

## Milestone 2 - Custom Object Creation

Six custom objects were created to form the data backbone of the system: **Customer** (stores customer personal and contact information), **Shipment** (manages parcel details, source, destination, and tracking information), **Delivery Agent** (maintains agent profiles and branch assignments), **Branch** (represents courier branch offices), **Delivery Update** (stores status updates for each shipment), and **Invoice** (manages billing details for delivered shipments). Each object was configured with appropriate record-naming conventions - including auto-generated tracking numbers for Shipment (SHP-00000 format), update numbers for Delivery Update (UPD-0000 format), and invoice numbers for Invoice (INV-0000 format) - along with reporting and search capabilities enabled.

## Milestone 3 - Tab Creation

To make each object accessible through the Salesforce navigation interface, a custom tab was created for all six objects. This allows end users to view, create, and manage records directly through the application’s menu, rather than relying on backend configuration screens.

## Milestone 4 - Fields & Relationships

A total of 34 custom fields were designed and implemented across the six objects, capturing all data required to run courier operations - customer contact and address details, shipment weight, price, and delivery type, agent status and joining dates, branch codes and locations, delivery update remarks, and invoice payment status. In addition, five relationships were established to connect the objects into a coherent data model: four Lookup relationships (Shipment → Customer, Shipment → Delivery Agent, Delivery Agent → Branch, and Invoice → Shipment) and one Master-Detail relationship (Delivery Update → Shipment), ensuring that every delivery update is permanently and structurally tied to its parent shipment record.

## Milestone 5 - Validation Rules

Nine validation rules were implemented to safeguard data quality across the system. On the Customer object, rules ensure that email addresses are never left blank, phone numbers contain exactly 10 digits, and pincodes contain exactly 6 digits. On the Shipment object, rules prevent zero or negative weight and price values, and block expected delivery dates set in the past. On the Delivery Agent object, a rule validates that the email address contains a proper “@” symbol. On the Branch object, a rule ensures the branch code is never left blank. Finally, on the Delivery Update object, a rule requires that

remarks be provided whenever a shipment’s status is marked “Delayed” - ensuring accountability and traceability for any delivery exceptions.

## Milestone 6 - Page Layouts

Each object’s page layout was customized to present information in a clear, sectioned format. For example, the Customer layout was divided into “Customer Information” (name, type, email, phone, active status) and “Address Details” (address, city, state, pincode) sections, while the Shipment layout was organized into “Shipment Details,” “Address Information,” and “Delivery Information” sections. This restructuring significantly improves usability by grouping related fields together and reducing visual clutter for end users.

## Milestone 7 - Flow Automation

To eliminate manual status synchronization, a Record-Triggered Flow named **Auto_Update_Shipment_Status** was built. This flow triggers automatically whenever a new Delivery Update record is created, and it updates the Status field on the corresponding parent Shipment record to match the status entered on the Delivery Update. This automation directly fulfills the project’s goal of real-time, error-free status tracking without requiring any manual intervention from staff.

## Milestone 8 - Lightning App

All six custom objects, along with Reports and Dashboards, were assembled into a single, cohesive Lightning Application named “Courier Management System.” This provides users with one central entry point to access every module of the system, streamlining daily navigation and operations.

## Milestone 9 - Profiles

Four custom profiles were designed to enforce the principle of least privilege across the organization. **Courier Admin** (cloned from System Administrator) has full system control. **Branch Manager** (cloned from Standard Platform User) can create, read, and edit Customers, Shipments, and Delivery Updates, with read/edit access to Agents and Branches, and read-only access to Invoices. **Delivery Agent** has read/edit access to Shipments and full access to Delivery Updates, with read-only visibility into Customers and Invoices. **Customer Support** has read/edit access to Customers and Shipments, with read-only access to Delivery Updates and Invoices. Session timeout and password policies were also configured for each profile to strengthen system security.

## Milestone 10 - Roles

A four-level role hierarchy was constructed to govern data visibility: **Courier Admin** sits at the top with organization-wide access, followed by **Branch Manager**, who oversees branch-level operations, and finally **Delivery Agent** and **Customer Support**, who each have

access limited to their assigned scope of work. This hierarchy ensures that sensitive shipment and billing data is only visible to users who genuinely need it.

## Milestone 11 - Users

Four users were provisioned, one for each role and profile combination - Courier Admin, Branch Manager, Delivery Agent, and Customer Support. Given the Developer Edition's limitation of two concurrent "Salesforce" licenses, the Courier Admin and Branch Manager users were assigned the standard Salesforce license, while the Delivery Agent and Customer Support users were assigned the Salesforce Platform license, ensuring the system remained within licensing constraints while preserving full functional access for each role.

## Milestone 15 - Reports & Dashboards

Three custom report types were created (Shipment Tracking Report, Customer Summary Report, and Agent Performance Report), forming the foundation for three operational reports: **Shipment Status Summary** (grouped by status, summarizing revenue), **Agent Performance Report** (grouped by delivery agent, showing shipment counts), and **Revenue by Branch Report** (grouped by branch and month, incorporating invoice payment status). These reports were then consolidated into a single **Courier Management Dashboard**, featuring a Donut Chart for shipment status distribution, a Bar Chart for agent performance comparison, and a Column Chart for branch-wise revenue trends - giving management a real-time, at-a-glance view of overall business performance. The dashboard was shared with the Courier Admin, Branch Managers, and Customer Support team to ensure appropriate visibility.

## Testing

Comprehensive functional testing was carried out to validate the entire system. Test records were created for each object to confirm that validation rules correctly blocked invalid data entries - such as improperly formatted phone numbers, negative shipment weights, and delayed statuses without remarks. The Flow automation was tested by creating Delivery Update records and confirming that the parent Shipment's status updated automatically and accurately. Finally, all reports and the dashboard were refreshed and verified to ensure they displayed correct, up-to-date data. Screenshots were captured throughout this process to document system reliability and correctness.

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# Conclusion

The Courier Management System successfully demonstrates how Salesforce's declarative automation tools - custom objects, validation rules, record-triggered flows, role-based security, and dashboards - can be combined to digitize and streamline a real-world logistics operation. What began as a set of disconnected, manual processes prone to delays and errors has been transformed into a centralized, real-time platform that improves operational efficiency, strengthens

data accuracy, and significantly enhances customer satisfaction. The role-based security model ensures that sensitive shipment and billing information remains protected while still enabling every stakeholder - from branch managers to delivery agents to support staff - to work efficiently within their scope of responsibility. Beyond meeting its immediate objectives, this project provided the team with practical, hands-on experience in core Salesforce administration concepts, including data modeling, process automation, and business intelligence reporting - skills directly transferable to real-world enterprise CRM implementations.

Looking ahead, the system has clear potential for further enhancement. Planned future scope includes a mobile tracking application for customers and agents, GPS integration for live parcel location tracking, SMS-based delivery notifications, and AI-driven delivery route optimization - each of which would further reduce manual effort and elevate the platform from an internal operations tool into a fully customer-facing logistics solution.